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Production of Cumene

Memo 1

Section 02

Group 03

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Chapter 1: Literature Search and Background

1.1 Process Background - Cumene Production

1.1.1 Product Description and Uses

Major Industrial Applications

Cumene, or Isopropyl benzene, (C_9H_{12}) is a colorless, flammable, aromatic hydrocarbon. Cumene is made primarily by the alkylation of benzene with propylene. Cumene is used industrially as a chemical intermediate, not as a product for direct use. New methods of producing cumene involve vapor-phase alkylation with a solid acid catalyst, a technique that is highly selective and environmentally friendly compared to the older aluminum chloride method ^[1] ^[2].

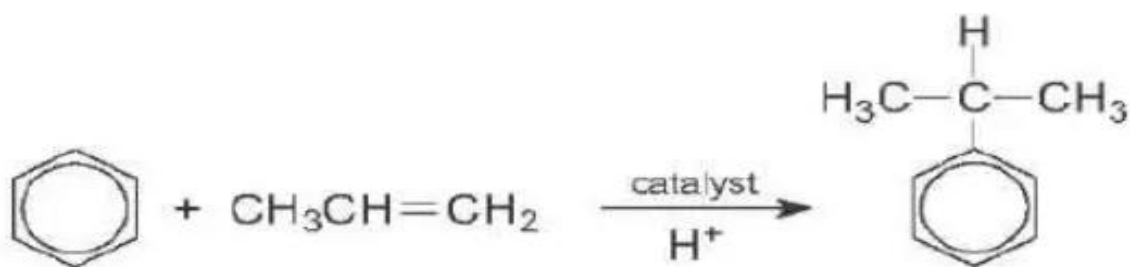


Figure 1: Cumene Reaction

The major use of cumene is in the production of phenol and acetone via the cumene (Hock) process. The cumene process involves the oxidation of cumene-to-cumene hydroperoxide, which is further treated with acid to yield phenol and acetone. It is estimated that around 90-95% of the total worldwide cumene production capacity is dedicated to the phenol and acetone industry ^[1].

Table 1: Industrial applications and uses for Phenol and Acetone ^[2]

Industrial applications and uses	Phenol	Acetone
	Polycarbonate plastics	Solvent applications
	Epoxy resins	Methyl methacrylate production
	Phenolic resins (adhesives, laminates)	Pharmaceuticals
	Caprolactam (Nylon-6 production)	Paints and coating

Primary Producers Worldwide and Annual Production Volumes (tons/year)

Similarly, the global cumene production is characterized by regional concentration, with the Asia-Pacific region accounting for more than 50% of the global capacity, making it the highest production region ^[4]. In the Asia-Pacific region, China is identified as the highest single-country producer of cumene, owing to the availability of extensive petrochemical infrastructure and strong downstream demand for phenol and acetone. North America, comprising the US, is identified as the second-highest producer of cumene, accounting for around 20-25% of the global capacity, primarily due to integrated refinery-petrochemical and phenolics production facilities. Europe is ranked as the third-highest producer of cumene, accounting for around 20% of the global capacity, with key manufacturing locations in Germany, the Netherlands, Belgium, and Spain. India, Japan, South Korea, Taiwan, and the Middle East account for the remaining 5-10% of the global capacity. ^[5]

Table 2: Production of cumene ^{[9] [10] [11] [12] [13]}

Company	Country	Approx. Cumene Capacity (tons/year)
INEOS Phenol	Europe	≈ 2,062,000 t/y
Shell plc	USA	≈ 600,000 – 800,000 t/y
ExxonMobil	USA / UK	≈ 600,000 – 900,000 t/y
BASF SE	Germany / Belgium / USA	≈ 500,000 – 800,000 t/y
SABIC	KSA	≈ 400,000 – 700,000 t/y

Feedstock and Product Market Prices (\$/ton)

The economics of cumene production are largely governed by the prices of two major feedstocks: benzene and propylene. The prices of these two feedstocks and, most importantly, the price of cumene are major factors governing the economics of production. Benzene and propylene are derived from crude oil refining and petrochemical production. The prices of these two major feedstocks have a major impact on the economics of production of cumene. The prices of cumene vary from region to region according to regional demand and supply and feedstock prices. For example, during late 2025 and early 2026, the spot prices of cumene varied from \$780 to over

\$950 per ton in major markets such as Asia, Europe, and North America. During the same period, the prices of benzene were largely ranging from \$700 to \$900 per ton, while propylene prices varied according to regional feedstock conditions.

Table 3: Market Prices of Feedstock and Product ^{[6] [7] [8]}

Material	Price Range (\$/ton)
Benzene (feedstock)	≈ 700 - 900
Propylene (feedstock)	≈ 800 – 1,100
Cumene (product)	≈ 780 - 950

Economic Incentives Driving Process Selection

The choice of technology in the production of cumene primarily depends on efficiency and cost of operation. Modern plants use zeolite-catalyzed liquid-phase alkylation instead of traditional aluminum chloride (AlCl_3)-based processes, because zeolite catalysts are more selective in forming cumene and less in forming undesired byproducts. The increase in selectivity reduces raw materials, separation, and recycle costs, thus increasing overall yields. As benzene and propylene are major contributors to variable costs, it appears that increasing efficiency in their utilization is an important objective. The zeolite-based process also eliminates corrosion and acidic waste streams, thus reducing maintenance costs. These costs are environmental compliance costs. ^{[1] [2]}

Energy efficiency and plant integration are major economic considerations. Distillations, including benzene recovery and cumene purification, are major energy-intensive processes, contributing to a significant portion of overall energy consumption. Modern plant designs incorporate improved heat integration and distillation configurations to minimize steam requirements and lower energy costs. Additionally, cumene plants are often integrated with downstream phenol and acetone plants, allowing for better feedstock efficiency, transportation savings, and energy efficiency. All these factors result in modern catalytic alkylation plants being the most economical option for commercial-scale production ^[3]

1.1.2 Typical Process Conditions

Cumene production on an industrial scale is accomplished by the vapor-phase alkylation of benzene with propylene using a solid acid catalyst in a fixed-bed reactor. The choice of operating conditions is a compromise between reaction kinetics, driving force, selectivity, and economic viability. Cumene production using modern catalysts based on zeolites is more active, selective, and environmentally friendly compared to the earlier liquid acid process.^[14]

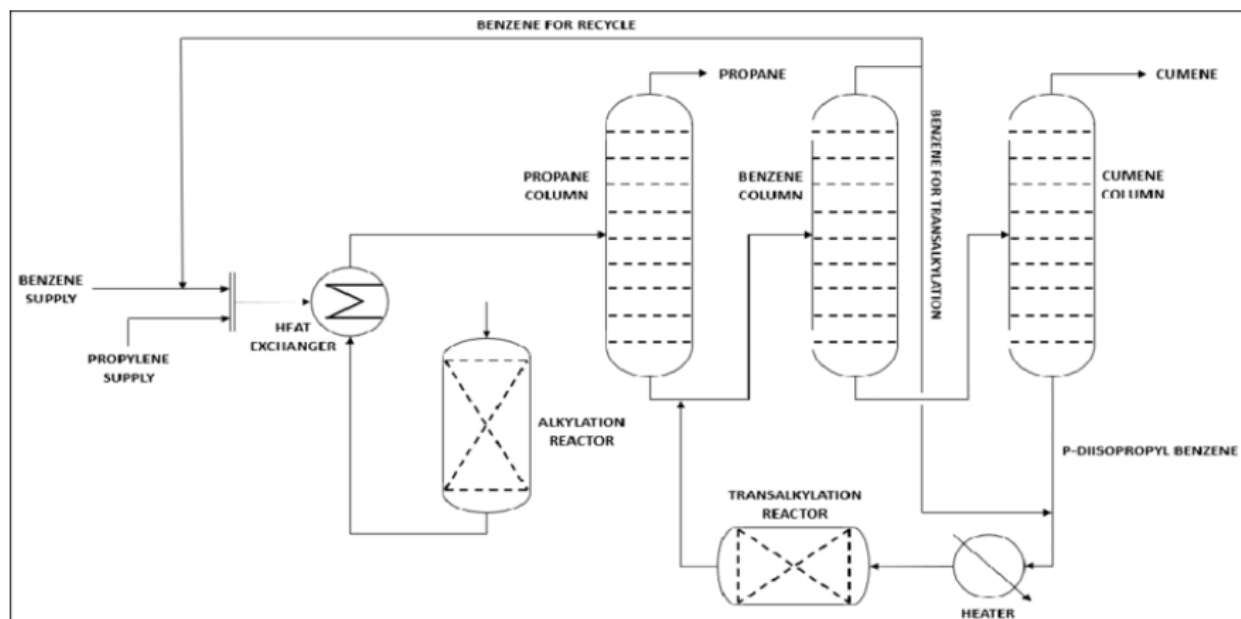


Figure 2: Cumene Process Flowchart

Feed Compositions

The reactor feed is dominated by benzene and propylene, with benzene being in large molar excess. The typical industrial ratio of benzene to propylene is between 5:1 and 10:1, although most commercial units approach a ratio of 10:1 before recycling. The addition of excess benzene prevents secondary alkylation reactions, especially the production of di-isopropyl benzene (DIPB), because it lowers the chances of propylene reacting with the cumene already present in the reactor. A lower concentration of cumene in the reactor improves selectivity and extends the life of the catalyst. The unreacted benzene is separated and recycled, leading to high overall plant yield despite moderate per-pass conversion.^[14]

In this design, the feed is supplied in two different streams. One stream is entirely 100% benzene, and the second stream is 95% propylene and 5% propane, where propane is considered an inert component. The addition of propane as an inert component to this feed stream is intended to help regulate the partial pressure of the propylene, as well as affect the overall composition of the reaction mixture, without participating in the alkylation process. This is also a more realistic representation of a feed stream, as a stream of propylene may contain small percentages of propane, which is considered an inert component, when it is separated from a refinery or a steam cracker unit.

Temperature Condition

The temperature range of 300 to 350 °C (573 to 623 K) is common in industrial alkylation reactors. Higher temperatures are required for economically feasible reaction rates, since the rate constant follows an exponential increase with temperature, as seen in Arrhenius behavior. The temperature range also helps in maintaining a vapor-phase environment, which is beneficial for mass transfer in the porous zeolite catalyst. Higher temperatures also lead to side reactions and catalyst deactivation due to coke deposition; thus, the chosen temperature range is a compromise between kinetic and selective factors. ^[14]

Pressure Condition

Typical operating pressures are maintained between 30 and 40 bar. Since the alkylation reaction is a process that reduces the total number of gas-phase moles (two moles of reactants to form one mole of product), the alkylation reaction is favored by higher pressure. Higher pressure also increases the partial pressure of the reactants, which increases the reaction rates and improves catalyst-reactant interaction in the dense vapor phase. ^[14]

Reactor Conversion

The conversion of propylene per pass through the reactor is generally restricted to around 30–40%. Although higher per-pass conversion might seem desirable, excessive conversion in a single pass increases the concentration of cumene within the reactor, thereby promoting secondary alkylation reactions such as the formation of di-isopropylbenzene (DIPB). Therefore, industrial operation deliberately maintains moderate per-pass conversion to preserve selectivity. However, through downstream separation and recycling of unreacted reactants, particularly benzene, the overall plant

conversion of propylene can reach approximately 99%. This strategy allows the process to achieve high overall efficiency while maintaining cumene selectivity above 95% under optimal operating conditions, ensuring effective utilization of propylene and minimizing byproduct formation. ^[14]

Reactor Type

The reactor design used is a vapor-phase Packed Bed Reactor (PBR) with a solid zeolite catalyst. The design allows for continuous processing, high selectivity, good temperature control, and minimized environmental effects compared to the traditional liquid acid process. The design is intended to provide single-phase vapor flow through the catalyst bed to avoid maldistribution, mass transfer limitations, and possible hot spots. ^[14]

1.1.3 Competing Feedstock Routes

Cumene production has traditionally and presently involved the alkylation of benzene with propylene. Various competing routes have existed based on the catalyst employed. These competing routes are summarized below:

1. Historic Homogeneous Acid Route (Liquid-Phase)

This traditional commercial process employs Friedel-Crafts alkylation in the presence of liquid acids such as sulfuric acid or Aluminium Chloride (AlCl_3) ^[15].

- **Advantages:** Relatively simple chemistry and well-established traditional industrial expertise ^[15].
- **Disadvantages:** The process is very corrosive, with difficult acid neutralization and toxic salt waste disposal ^[16]. Moreover, it is less selective for cumene than more contemporary processes, and this traditional process is therefore obsolete and no longer favoured for new commercial plants ^[15].

2. Solid Phosphoric Acid (SPA) Catalyst Route

This process uses a fixed-bed reactor with solid phosphoric acid supported on alumina or silica ^[17].

- **Advantages:** It does not involve the handling of liquid acid and has been shown to be successful on an industrial scale for the production of high purity cumene after distillation ^[17].

- **Disadvantages:** The catalyst is prone to deactivation by coking and is non-regenerable, making it very expensive to replace ^[16]. It also produces large amounts of poly-isopropyl benzenes (PIPBs) that require large trans alkylation units, and it is less energy efficient compared to newer processes ^[17].

3. Zeolite-Based Technologies (Modern Industrial Standard)

Modern processes (e.g., Badger, UOP Q-Max) use solid zeolite catalysts in gas or liquid-phase alkylation, usually with a separate trans alkylation reactor for converting PIPB to cumene ^[16, 17].

- **Advantages:** It provides a very high level of selectivity and propylene conversion, resulting in an overall purity of cumene of 99.9 wt.% ^[16]. The catalyst is regenerable and non-corrosive ^[17]. Moreover, it provides a high potential for energy integration (such as dividing-wall columns) and an excellent environmental profile ^[17].
- **Disadvantages:** This process has a higher initial capital investment (Capex) for catalyst and technology licensing, as well as a more complicated process flow sheet ^[17].

4. Emerging Route: Benzene + Propane

A new process in which propane is dehydrogenated to propylene using a Pt-alumina catalyst, and then benzene is alkylated in a single catalytic system ^[18].

- **Advantages:** This process makes the C₃ supply chain simpler by not requiring the purchase or production of propylene separately ^[18].
- **Disadvantages:** The process is still in the research stage and has low single-pass conversion. Its feasibility and reliability for a commercial plant with a capacity of 100,000 tons per year have not been established yet ^[18].

Technical and Economic Justification

For achieving the required production capacity of **100,000 tons/year**, the **Zeolite-based Route** is the most preferred and feasible option. The prevailing market prices make it imperative to be cost-effective in raw material usage (Benzene: \$1,000/ton, Propylene: \$950/ton, Cumene: ~\$1,350/ton) ^[19]. The nearly 100% yield and recycling of PIPB in the zeolite route make it highly advantageous in terms of minimizing costly feed material losses ^[2]. Additionally, the absence of corrosion

problems and extending the life of the catalyst make a substantial impact on lowering operating costs (OPEX), which is well within the additional Capex costs and makes it highly profitable [17, 19].

1.1.4: Reactor Types & Product Distribution

Common Reactors for Cumene Chemistry

The alkylation of benzene with propylene to produce Cumene is an exothermic reaction that relies on acid catalysis [20] [21]. While theoretical evaluations might consider Batch, Continuous Stirred Tank Reactors (CSTR), or Plug Flow Reactors (PFR), modern industrial Cumene production strictly utilizes continuous catalytic reactors [21]. Batch reactors are entirely unsuitable for a production capacity of 100,000 tons/year due to excessive downtime and high operating costs. While a CSTR could theoretically be used for liquid phase homogeneous catalysis (like older aluminum chloride processes), the industry has shifted away from highly corrosive, tough-to-separate liquid catalysts [20].

Selection of the Fixed-Bed Reactor (PBR)

A fixed bed reactor (PBR) is universally chosen for modern Cumene production. The process requires bringing the liquid or mixed-phase reactants into contact with a solid acid catalyst, typically a specialized zeolite. The PBR is optimal because:

- It allows for continuous, high-volume production without the need for catalyst separation units downstream.
- The catalyst remains stationary, preventing mechanical wear and attrition [20].
- The fixed-bed design accommodates multiple catalyst beds with interstage cooling (often achieved by injecting cold reactants between beds) to tightly manage the highly exothermic nature of the alkylation reaction, preventing catalyst degradation and thermal runaways [21].

Typical Byproduct Formation Patterns

In the alkylation reactor, the primary reaction is accompanied by several consecutive and side reactions. The most prominent byproduct formation pattern is polyalkylation. Once cumene is formed, its aromatic ring is more reactive to alkylation than the original benzene ring.

Consequently, propylene reacts with cumene to form Di-Isopropyl Benzene (DIPB), and to a lesser extent, Tri-Isopropyl Benzene (TIPB). Additionally, if pure propylene is not used, minor byproducts can include oligomers of propylene (like hexenes) and ethylbenzene (if ethylene impurities are present in the feed ^[21]).

Reaction Selectivity in Industry

To manage the polyalkylation byproducts, industrial processes operate with a high molar ratio of benzene to propylene to favor single alkylation ^[21]. Despite this, single-pass selectivity to cumene typically hovers around 85% to 90% ^[22]. However, modern fixed-bed zeolite processes incorporate a separate trans alkylation reactor ^[22]. In this unit, the separated DIPB and TIPB byproducts react with excess benzene to form additional cumene. By utilizing trans alkylation, the overall industrial process selectivity to cumene is exceptionally high, typically exceeding 99% ^[20].

1.1.5 Sustainability Consideration

Environmental Impact of Current Processes

One of the main impacts of cumene production on the environment is air pollution. Benzene, propylene, and cumene are volatile hydrocarbons that may be released as volatile organic compounds (VOCs). Since benzene is classified as a possible human carcinogen, it is particularly hazardous. Thus, to reduce the risks to the environment and human health, strict control of emissions is required.^[24]

Since the major reaction in the cumene process occurs in the hydrocarbon phase, water pollution is generally minimized. In contrast, emission control systems, condensate streams, and equipment cleaning may all generate wastewater. This wastewater must be treated prior to discharge because it may contain hydrocarbons. Prior to discharge into the environment, wastewater must be properly treated and its organic content reduced in accordance with environmental regulations. ^[25]

With the use of zeolite catalysts instead of more antiquated liquid acid catalysts such as aluminum chloride. The waste from older technologies was acidic and had to be neutralized before it could be disposed of. With modern zeolite catalysts being solid and regenerable, the problems of generating hazardous waste and corrosion are significantly diminished.^[26]

Another major environmental concern associated with this process is the use of energy. The process requires high energy input, especially in the distillation of cumene and benzene. This increases the emission of carbon dioxide (CO₂) and fuel consumption.^[27]

Energy Efficiency Metrics

Specific energy consumption (SEC), usually expressed in GJ per metric ton of product and factoring in the total fuel, steam, and electricity consumption in the alkylation and separation processes, is the key measure used to evaluate energy efficiency ratios in cumene production. Because of the separation-intensive nature of cumene production, especially in the benzene recovery and product purification steps, reboiler duty, condenser duty, and steam consumption per ton of product are also significant process measures. These measures directly correlate to the energy required for distillation. Heat recovery efficiency and energy intensity (total energy input divided by product output) are important measures of overall thermal efficiency at the process level. Because higher energy consumption translates to higher fuel consumption and CO₂ emissions, these measures are closely tied to environmental issues.^[28]

Waste Streams and disposal

Although there is relatively fewer waste streams associated with modern cumene production than in traditional liquid acid processes, there are still several important waste streams that require proper management and disposal. The major waste streams are: wastewater containing dissolved hydrocarbons, used solid catalysts, and vent gases containing volatile organic compounds (VOCs) such as cumene and benzene. In order to reduce the emission of volatile organic compounds, gas emissions are typically treated by thermal oxidation, flaring, or recovery systems. Condensate, equipment washes, and separation unit wastewater is treated by oil-water separation, followed by physicochemical or biological treatment, and then released. Spent zeolite catalysts, which are recycled, regenerated, or disposed of in accordance with hazardous waste regulations, comprise the largest portion of solid waste. To reduce environmental impact and ensure regulatory compliance, modern Best Available Techniques (BAT) emphasize waste minimization, hydrocarbon recovery, leak detection and repair (LDAR) programs, and proper catalyst management.^[26]

Emerging “Green” Alternatives

The new "green" cumene production methods focus on minimizing energy consumption, reducing dependence on fossil resources, and improving the sustainability of catalysts. A breakthrough in this area is the continuous development of high-selectivity zeolite catalysts, which minimize the energy required for waste separation and production by reducing the number of byproducts. To minimize the carbon footprint associated with cumene production, research is also being conducted on the use of biobased propylene derived from bioethanol or biomass-based routes to partially substitute petroleum-based resources. In addition, advanced heat integration and process intensification techniques are being employed to minimize fuel and steam consumption in distillation columns. To improve catalyst lifespan and minimize solid waste, some research is also being conducted on more environmentally friendly alkylation processes and improved catalyst regeneration technology. These approaches aim to minimize greenhouse gas emissions, maximize resource efficiency, and improve the sustainability of cumene production processes while maintaining productivity. [27]

1. 2 Reaction Kinetics of the Cumene Reaction

The cumene ($C_3H_6 + C_6H_6 \rightarrow C_9H_{12}$) process is the primary industrial method for producing phenol and acetone. The method consists of three main steps: alkylation of benzene with propylene, acid-catalyzed breakdown of cumene hydroperoxide (CHP), and oxidation of cumene to CHP. Accurate kinetic models are essential for reactor design and optimization [31].

1.2.1 Mechanisms of the Cumene Process

The cumene process follows several primary reaction mechanisms, which dictate kinetic behavior and catalyst selection:

- **Alkylation (Benzene + Propylene):** Occurs over solid acid catalysts zeolite. The mechanism follows a Langmuir-Hinshelwood (L-H) model where benzene and propylene adsorb on the catalyst's acid sites [31].
- **Oxidation (Cumene to CHP):** A liquid-phase Radical Chain Mechanism involving initiation, propagation, and termination steps. In the absence of a catalyst, this process requires significant thermal initiation [33].

- **Decomposition (CHP to Phenol/Acetone):** Characterized by an Autocatalytic Mechanism during thermal decomposition, where the conversion of CHP accelerates the reaction rate through radical intermediates ^[34].

1.2.2 Kinetic Models and Rate Expressions

Several kinetic models have been proposed for the cumene process, depending on the catalyst type and reaction conditions. The most widely accepted models include Langmuir-Hinshelwood, Power-Law, and Autocatalytic mechanisms.

- **Alkylation and Trans alkylation Models:**

Trans alkylation of ,1,4-diisopropylbenzene (DIPB) with benzene follows a Langmuir Hinshelwood expression, which can be simplified to a second-order surface reaction:

Equation 1: Trans alkylation Expression LH ^[31]

$$r = \frac{kK_B K_{DIPB} (C_B C_{DIPB} - \frac{C_C}{K_{eq}})}{(1 + K_B C_B + K_{DIPB} C_{DIPB} + K_C C_C)^2}$$

Alternatively, empirical power-law models describe the main alkylation step:

Equation 2: Power-Law Alkylation ^[32]

$$r = k C_B^a C_P^b$$

- **Oxidation and Decomposition Models:**

Cumene oxidation follows an apparent first-order model for the initiation phase

Equation 3: Apparent First-Order Oxidation ^[33]

$$r = k C_{\text{Cumene}}$$

The thermal decomposition of CHP is characterized by an Autocatalytic model:

Equation 4: Autocatalytic Decomposition ^[34]

$$\frac{d\alpha}{dt} = k \alpha^m (1 - \alpha)^n$$

Where:

- α = fractional conversion of CHP
- k = rate constant
- m, n = reaction order parameters

Typical values reported:

$$m = 0.5, n = 1.5$$

This nonlinear behavior reflects the accelerating reaction rate as decomposition proceeds, which is critical for thermal safety and runaway analysis.

Table 4 : Kinetic Expressions and Reaction Orders for Cumene Process

Model	Expression	Reaction Order (Reactants)	Reference
Langmuir–Hinshelwood	$r = k C_B C_{DIPB}$	Order 1 (B), Order 1 (DIPB)	Thakur et al. (2016)
Power-Law	$r = k P_B P_P$	Order 1 (B), Order 1 (P)	Sridevi et al. (2001)
Apparent First-Order	$r = k C_{\text{Cumene}}$	Order 1 (Cumene), Order 0 (O ₂)	Krongauz et al. (2014)
Autocatalytic	$\frac{d\alpha}{dt} = k \alpha^m (1 - \alpha)^n$	m = 0.5, n = 1.5 (Overall 2)	Duh (2016)

1.2.3 Kinetic Parameters and Arrhenius Activation Energy

The temperature dependence of the rate constants follows the Arrhenius Relationship. A summary of the activation energy (E_a), pre-exponential factor (k_0), and valid temperature ranges is provided in Table 2.

Table 5 Kinetic Parameters for Cumene Process Reactions

Reaction Step	E_a (kJ/mol)	k_o (Units)	Temp Range
Trans alkylation	116.53	2.14×10^{11}	493–593 K
Alkylation (Main)	55.4	k_{avg}	180–240 °C
Oxidation (Thermal)	120.6	$1.55 \times 10^{-9} \text{ s}^{-1}$	70–130 °C
CHP Decomposition	124.5	$1.05 \times 10^{13} \text{ min}^{-1}$	80–150 °C

2. Cumene Production Calculation

Objective

This section will outline the mass balance of the system to produce cumene, and this process is considered an alkylation of benzene with propylene. The aim of this calculation is to find out the rates of all streams entering and exiting the reactor, considering a specified propylene conversion and selectivity in the production of cumene.

Some assumptions and considerations that we should consider:

- Steady state
- Negligible pressure
- Isothermal system
- Exothermic reaction
- Irreversible reaction
- The conversion of propylene is 0.99
- Variable volume
- Heterogeneous gas phase with solid catalysts
- Shutdown of 2 weeks per year (50 weeks remain)
- Physical and chemical properties remain constant

Molecular weights:

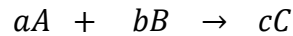
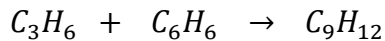
C_3H_6 (Propylene) = 42.081 kg/kmol

C_6H_6 (Benzene) = 78.114 kg/kmol

C_9H_{12} (Cumene) = 120.195 kg/kmol

C_3H_8 (Propane) = 44.097 kg/kmol

2.1 Mass Balance Calculation



The ratio of the coefficient on stoichiometric equation is 1:1:1 so: $b/a = 1$, $c/a = 1$

The Overall mass Balance Equation

$$F_{A0} - F_A + \int_0^w r_A dV = \frac{dN_a}{dt}$$

Steady state so $\frac{dN_a}{dt} = 0$

$$F_{A0} - F_A + \int_0^w r_a' dW = 0$$

$$\int_{F_{A0}}^{F_A} \frac{dF_A}{r_a'} = \int_0^w dW$$

$$conversion(x) = \frac{F_{A0} - F_A}{F_{A0}}$$

$$X \cdot F_{A0} = F_{A0} - F_A$$

$$dF_A = -F_{A0} dx$$

At F_{A0} , $x = 0$

At F_A , $x = x$

$$\int_0^x \frac{dF_A}{r_a'} = \int_0^w dW$$

The sample calculation will be shown in Appendix A

The feed rate F_C conversion from tons/year to kmol/s

$$F_C = \frac{100,000 \text{ tons}}{1 \text{ year}} \times \frac{1 \text{ year}}{50 \text{ weeks}} \times \frac{1 \text{ week}}{7 \text{ days}} \times \frac{1 \text{ day}}{86,400 \text{ sec}} \times \frac{1000 \text{ kg}}{1 \text{ ton}} \times \frac{\text{kmol}}{120.19 \text{ kg}}$$

$$F_C = 0.0275 \frac{\text{kmol}}{\text{s}} \approx 99.049 \frac{\text{kmol}}{\text{h}}$$

Table 6: Mass Balance Results

Species	Inlet Molar flow rate (kmol/s)	Outlet Molar flow rate (kmol/s)	Inlet Mass flow rate (kg/s)	Outlet Mass flow rate (kg/s)
propylene	0.0278	0.000278	1.170	0.0323
benzene	0.0292	0.000292	2.281	0.0228
cumene	0	0.0275	0	3.305
propane	0.00146	0.00146	0.064	0.064
Total	0.05846	0.02953	3.515	3.424

Table 7: Stoichiometric Table for Flow System

Species	Inlet	change	remaining
propylene	F_{A0}	$-F_{A0}X$	$F_{A0}-F_{A0}X$
benzene	F_{B0}	$-b/aF_{A0}X$	$F_{B0}-b/aF_{A0}X$
cumene	F_{C0}	$+c/aF_{A0}X$	$F_{C0}+c/aF_{A0}X$
propane	F_{I0}	0	F_{I0}
Total	F_T		$F_T+(c/a-b/a-1) F_{A0}X$

Some equations that will be used

$$\delta = \left(\frac{c}{a} - \frac{b}{a} - 1\right)$$

$$\varepsilon = \delta y_{Ai}$$

$$\theta_i = \frac{F_{i0}}{F_{A0}} = \frac{C_{i0}}{C_{A0}} = \frac{y_{i0}}{y_{A0}}$$

$$y_{i0} = \frac{F_{i0}}{F_{A0}} = \frac{C_{i0}}{C_{A0}}$$

Table 8:Stoichiometric for cumene product

Species	Feed rate to reaction (mol/s)	Change within reactor (mol/s)	Effluent rate from reactor (mol/s)
Propylene (A)	27.750	-27.473	0.278
Benzene (B)	30.750	-27.473	3.278
Cumene (C)	0	27.473	27.473
Propane (Inert)	1.460	0	1.46
Total	59.960	-27.473	32.488

2.3 Python Code

Submitted as a separate file.

```
=====
FOGLER STOICHIOMETRIC TABLE [X = 0.9900]
=====
F_i = F_i0 + nu_i * F_A0 * X
=====
```

Species	nu_i	F_i0 (mol/s)	Change (mol/s)	F_i (mol/s)	y_i	m_i (kg/s)	w_i
Propylene (C3H6)	-1.00000	27.77778	-27.50000	0.27778	0.00897	0.01167	0.00332
Benzene (C6H6)	-1.00000	29.23977	-27.50000	1.73977	0.05616	0.13570	0.03864
Cumene (C9H12)	1.00000	0.00000	27.50000	27.50000	0.88768	3.30000	0.93972
Propane (C3H8)	0.00000	1.46199	0.00000	1.46199	0.04719	0.06433	0.01832
TOTAL		58.47953	-27.50000	30.97953	1.00000	3.51170	1.00000

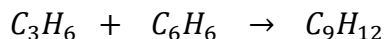
```
=====
STREAM SUMMARY [X = 0.9900]
=====
Limiting reactant : Propylene (C3H6)
Conversion : X = 0.9900
Total molar flow : F_total = 30.97953 mol/s
Total mass flow : m_total = 3.51170 kg/s
=====
```

Figure 3:Python Result

2.4 Heat of Reaction

The aim is to quantify the enthalpy change associated with the alkylation of benzene. This is vital in the sizing of cooling systems in reactors, providing a safe temperature environment, and comparing the energy demands of the conventional propylene route with the new propane route.

Heat of reaction calculated at 298 K



$$\Delta H_{rxn}(T_r) = \Sigma H_{f(product)} - \Sigma H_{f(reactant)}$$

$$H_{Benzene} = 82.9 \text{ kJ/mol}^{[38]}$$

$$H_{Propylene} = 20.4 \text{ kJ/mol}^{[38]}$$

$$H_{Cumene} = 3.9 \text{ kJ/mol}^{[38]}$$

$$\Delta H_{rxn} = (3.9) - (82.9 - 20.4)$$

$$\Delta H_{rxn} = -99.4 \text{ kJ/mol}$$

Justification

The value of $\Delta H_{rxn} = -99.4 \text{ kJ/mol}$ shows that the alkylation of benzene with propylene is a highly exothermic reaction. In terms of its application to the industrial scenario, it shows that for every mole of cumene formed, 99.4 kJ of energy will be released to the system. This value, as stated under the project guidelines, will be important for the proper sizing of the cooling equipment to be used for the reactor to prevent runaway reactions and to maintain the integrity of the catalyst used for the process, which is zeolite.

We will calculate the heat of reaction for the Alternative feedstock:

$$\Delta H_{rxn}(T_r) = [\Delta H_{cumene}^{\circ} + \Delta H_{H_2}^{\circ}] - [\Delta H_{Benzene}^{\circ} + \Delta H_{propane}^{\circ}]$$

$$\Delta H_{rxn} = [3.9 + 0] - [82.9 + (-103.8)]$$

$$= +24.8 \text{ kJ/mol}$$

Comment on the Differences Between Feedstocks

The above vapor phase calculations highlight the differences between the two routes as follows:

1. Exothermic Advantage of Route 1: The Propylene route, which is the main route, is more exothermic as its $\Delta H_{rxn} = -99.0 \text{ kJ/mol}$. This would cause the temperature to increase as the reaction progresses in a fixed bed reactor. To achieve an isothermal condition for the plant

producing 100,000 tons/year, a multi-tubular reactor design could be employed to minimize the exothermic heat generated in the reaction process. [36, 37]

2. Endothermic Limitation of Route 2: The propane route, which is endothermic as its $\Delta H_{rxn} = +24.8$ kJ/mol, would not be self-sustaining as a reaction process. This process would require a constant input of energy to sustain the reaction. Moreover, the energy level for the Propylene route is nearly four times greater in comparison to the propane route as indicated by the absolute values of their respective ΔH_{rxn} values, i.e., $(99.0 > 24.8)$. [37]

3. Process Implications: The reaction process being in the vapor phase indicates that the high exothermicity of the Propylene route could cause a sintering problem for the catalyst. [36]

2.5 Energy Balance

Isothermal process

The temperature of isothermal process remains constant, where the temperature of reactants equal to the temperature of the products

$$Q = F_{A0} \Delta H_{rxn} X$$

$$Q = 27.75 \frac{\text{mol}}{\text{s}} * 0.99 * \left(-99.4 \frac{\text{KJ}}{\text{mol}} \right) = -2730.76 \text{ KW}$$

Overall Adiabatic Energy Balance

If the reactor is adiabatic, then no heat is removed from the reactor ($\dot{Q} = 0$) and the energy produced due to the reaction is used to increase the temperature of the gas mixture. The outlet temperature (T) can be calculated using the adiabatic energy balance equation:

$$T_{out} = T_0 - \frac{Q}{C_{p_{mix}} F_T}$$

Using Shomate equation to calculate C_p

$$C_{p_{mix}}(T) = A + Bt + Ct^2 + Dt^3 + \frac{E}{t^2}$$

Let $t = T/1000$

Table 9: Shomate constant table

Compound	Benzene	Propylene	Propane
A	1.4470	1.2111	-26.0099
B	391.8570	234.2472	361.6434
C	-232.5060	-107.6020	209.3942
D	52.5139	16.9831	47.6985
E	-1.6379	0.2383	0.8905
$C_{p_{mix}}$ J/mol·K	160	108	128

Sample calculation

$$C_{p_{mix}}(T) = 1.4470 + 391.8570t - 232.5060t^2 + 52.5139t^3 - \frac{1.6379}{t^2} = 160 \text{ J/mol} \cdot \text{K}$$

$$C_{p_{mix}} = \sum y_i C_{p_i}$$

C_p Values at inlet temperature (325°C) ^[38]

C_p (benzene) = 160 J/mol·K

C_p (propane) = 128.7 J/mol·K

C_p (propylene) = 108.00 J/mol·K

$$C_{p_{mix}} = 0.5 * 160 + 0.025 * 128.7 + 108.00 * 0.475 = 134.52 \text{ J/mol} \cdot \text{K}$$

$$T_{out} = 598.15 - \frac{-2730.76 \text{ KW}}{134.52 * 59.96} = 936.71 \text{ K}$$

APPENDIX A

Hand Calculation

$$T = 325^{\circ}\text{C} \quad \text{and} \quad P = 35 \text{ bar}$$

The Composition of reactant

$$y_A \rightarrow \text{Propylene} \rightarrow \text{limiting reagent} = 0.475$$

$$y_B \rightarrow \text{Benzene} \rightarrow \text{Excess reagent} = 0.5$$

$$y_C \rightarrow \text{Cumene} \rightarrow \text{product} = 0$$

$$y_I \rightarrow \text{Inert} = 0.025$$

$$C_{A_0} = y_{A_0} \frac{P}{RT} = 0.475 \frac{35 \times 10^5}{8314 \times 598} = 0.33882 \text{ mol/dm}^3$$

$$C_{A_0} = 0.33882 \text{ mol/dm}^3 = 338.82 \frac{\text{mol}}{\text{m}^3}$$

$$\delta = \left(\frac{c}{a} - b/a - 1 \right) = (1 - 1 - 1) = -1$$

$$\varepsilon = \delta \lambda_{A_0} = -1 (0.475) = -0.475$$

$$C_A = 338.82 \frac{(1 - 0.99)}{(1 - 0.475 \times 0.99)} = 6.395 \text{ mol/m}^3$$

$$F_C = 100,000 \frac{\text{ton}}{\text{year}} = 27.5 \frac{\text{mol}}{\text{s}}$$

$$C_C = C_{A_0} \frac{x}{(1 - 0.475 \times 0.99)} = 633.18 \text{ mol/m}^3$$

$$V = \frac{F_C}{C_C} = \frac{27.5}{633.18} = 0.043 \text{ m}^3$$

$$V_0 = \frac{0.0434}{(1 - 0.475 \times 0.99)} = 0.0819 \text{ m}^3$$

$$F_{A_0} = C_{A_0} \times V_0 = 338.82 \times 0.0819 = 27.75 \text{ mol/s}$$

$$F_A = C_A \times V = 6.395 \times 0.0434 = 0.2775 \text{ mol/s}$$

$$F_B = F_{A0} \frac{\theta_B + b/a X}{(1 + \epsilon X)}$$

$$= 27.75 \times \frac{0.5}{0.475} - 0.99}{(1 - 0.475 \times 0.99)} = \boxed{3.28 \text{ mol/s}}$$

$$F_B = F_{B0} - b/a F_{A0} X$$

$$3.28 = F_{B0} - 1(27.75)(0.99)$$

$$F_{B0} = \boxed{30.75 \text{ mol/s}}$$

$$F_{C0} = 0$$

$$F_C = F_{C0} + c/a F_{A0} X$$

$$F_C = 0 + 1(27.75)(0.99)$$

$$F_C = \boxed{27.5 \text{ mol/s}}$$

$$\frac{F_{I0}}{F_{A0}} = \frac{y_{I0}}{y_{A0}} \Rightarrow \frac{F_{I0}}{27.75} = \frac{0.025}{0.475}$$

$$F_{I0} = \boxed{1.46 \text{ mol/s}}$$

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